AI Messaging Sales Assistant

Planning document: WhatsApp, Instagram and Facebook Messenger, with n8n automation

Version 4.0 | Self-service platform for all industries | Status: plan for approval, no code built yet

1. Goal

Answer every customer message on WhatsApp, Instagram and Facebook Messenger within seconds, 24 hours a day, in Chinese and English. The assistant captures leads, books appointments, and hands the conversation to a staff member whenever a human is needed. It must be easy for three groups of people:

Who	What easy means for them
Customer	Instant reply in the chat app they already use, in their own language. A person is always one message away.
Staff	One inbox for all three channels, an AI summary on every handover, and a one-word command to give the chat back to the bot.
Business owner	A password-protected admin panel to change prices, FAQs and AI behaviour without a developer, a daily summary on the phone, and leads and bookings in one place.
New business (self-service)	Signs up alone, describes the business, and has a working AI assistant the same day, with no developer and no waiting for you.
Platform owner (you)	See every business, its usage and revenue, keep data separate, and pause any business that breaks the rules.

2. Decisions already made

Topic	Decision
Scope	**Multi-business platform for all industries**, each business fully separated (sections 5d and 5e).
Business model	**Self-service sign-up**: businesses register, set up, and pay by themselves. Invite-only beta first, then open sign-up.
WhatsApp and Meta	Each business uses its own WhatsApp Business number and Meta account. Guided manual connect first, one-click connect later.
Industries	**Any industry**: a universal base, a library of starter templates, and AI-assisted setup (section 5e). Aesthetic clinic is the first template and demo.
Languages	Chinese (Simplified and Traditional) and English.
First channel	WhatsApp. Instagram and Messenger follow using the same workflows.
Hosting	Free hosting for development and the invite-only beta. **Open public sign-up needs a paid server** (sections 4 and 5f).
Automation	n8n (self-hosted, free community edition).
Unified inbox	Chatwoot (open source, self-hosted).
Product scope	The functions of a SaleSmartly-style system (Chat, Customers, Marketing, Workflow, Reports, Team, Settings) **plus extra smart and convenient features** (section 5c).
Build approach	Assemble open source first (Chatwoot + n8n + our admin panel). Merge the screens into one interface later only if staff want it.
Admin system	Custom admin panel (web page, password-protected) inside the AI service.
Calendar	Google Calendar, two-way, **one calendar per practitioner**.

3. Recommended architecture

Four free open-source parts plus one AI service. Chatwoot gives staff the inbox and the handover feature, so none of that has to be built.

Layer	Tool	Role
Channels	Meta WhatsApp Cloud API, Instagram and Messenger APIs	Customer messages arrive from all three through one Meta app.
Inbox	Chatwoot (self-hosted)	Unified inbox, staff reply screen and mobile app, labels, assignment, conversation status.
Automation	n8n (self-hosted)	Routes messages, calls the AI, sends replies, runs handover alerts, reminders and the daily report.
AI service	Small Python service (FastAPI) in the aichat repo	Holds the prompt, the clinic knowledge, and the tools: availability, booking, lead save, handover.
AI model	Claude via the Anthropic API	Understands Chinese and English, follows the rules, decides which tool to use.
Admin panel	Web pages in the same AI service	Owner and staff manage settings, services, FAQ, leads and bookings, and test the bot.
Calendar	Google Calendar API (free)	Source of busy time and home of every booking, one calendar per practitioner.
Sign-up and billing	Public website, sign-up flow, payment provider such as Stripe	Businesses register, pick a plan, pay, and see their usage. Plans set limits.
Database	PostgreSQL	Settings, conversations, leads, bookings, reminders. Also used by Chatwoot and n8n.

Message flow

```
Customer (WhatsApp / Instagram / Messenger)
        |
        v
   Meta webhook ---> Chatwoot (inbox)
                        |  webhook
                        v
               n8n: Inbound Router
                  |- staff already handling?  -> stay silent
                  |- call AI service /respond
                  |      |- tool: check_availability (Google Calendar)
                  |      |- tool: create_booking (writes calendar event)
                  |      |- tool: save_lead (score hot/warm/cold)
                  |      '- tool: handover_to_human
                  |- send reply through Chatwoot -> customer (target: a few seconds)
                  '- if handover: private note + open conversation + alert staff
```

4. Free hosting plan

Scope of free hosting: good for development, the demo clinic and an invite-only beta with a few businesses. Open public sign-up needs a stable paid server, backups and monitoring (section 5f).

Everything runs in Docker Compose on one machine. The only public address needed is Chatwoot, because Meta must reach its webhook. n8n and the AI service stay private.

Option	Cost	Good for	Watch out for
Always-free cloud VM (for example Oracle Cloud Always Free)	$0	True 24/7 running. Best fit for the goal.	Needs a card for identity check. Free capacity varies by region. Setup takes longer.
Your own PC or spare machine at the office	$0 (electri city)	Testing and pilot.	Must stay on 24/7. Needs 8 GB RAM recommended (Chatwoot is heavy), 4 GB minimum.
Public address: ngrok free static domain	$0	Stable webhook URL that does not change.	One free domain per account. Free-tier limits apply.
Public address: Cloudflare quick tunnel	$0	No account needed.	URL changes on every restart, so Meta webhook must be re-entered. Not for production.

Honest limits of free: a laptop that sleeps cannot answer messages at 3 a.m. For real 24/7 operation use an always-on machine. Also, the AI model is the one component that is not free to run (see section 11), but it is very cheap at this volume.

5. n8n workflows

#	Workflow	Trigger	What it does
1	Inbound router	Chatwoot webhook (new message)	Ignores staff and private notes, calls the AI service, posts the reply, runs handover steps. A private note with the text /bot from staff gives the chat back to the bot.
2	Booking reminders	Every 15 minutes	Asks the AI service which bookings are due, sends a reminder 24 hours and 2 hours before. Uses a WhatsApp template if the 24-hour window is closed.
3	Daily report	Every day, 9:00	Sends a summary to the owner on Telegram: new chats, leads by score, bookings, handovers, tomorrow's appointments.
4 (later)	Lead follow-up	Scheduled	Nudges hot leads who did not book, inside the allowed message window.

5a. Admin panel

A small password-protected web page, served by the AI service on the private network and reached through a secure link. Everything the owner would otherwise ask a developer to change is here. Changes apply on the next customer message, with no restart.

Page	What the owner or staff can do
Dashboard	Chats today, leads by hot / warm / cold, bookings, handovers waiting, bot status.
AI settings	Bot on or off (all channels or per channel), greeting, tone, handover rules, languages, after-hours message.
Services and prices	Add, edit or hide treatments with English and Chinese names, duration, price, and which practitioners can perform each one.
Practitioners	Add or remove practitioners, working days and hours, linked Google Calendar.
Hours and closures	Clinic opening hours and holiday dates.
Knowledge / FAQ	Edit bilingual answers, policies and promotions in a simple editor.
Leads	List, filter, mark status, export to CSV.
Bookings	View, cancel, reschedule, mark attended or no-show. Changes sync to Google Calendar.
Test chat	Talk to the bot in English or Chinese using the saved settings before going live.

Page	What the owner or staff can do
Staff alerts	Telegram or email contacts for handovers and the daily report.
Users	Owner and staff logins with roles. Staff can manage bookings and leads, only the owner can change AI settings.

5b. Google Calendar, one per practitioner

The assistant never double-books and never offers a time a practitioner has blocked.

Part	How it works
Availability	For a service, the bot looks at the calendars of the practitioners who can do it, combines clinic hours, treatment length and existing events, and offers the real open slots. If a practitioner blocks time in their own calendar, it disappears from the offer.
Booking	The booking is saved and a calendar event is created on the chosen practitioner's calendar with customer name, phone, service and channel. Rechecked just before saving so two customers cannot take the same slot.
Change or cancel	Cancelling or rescheduling from chat or the admin panel updates or removes the event.
Practitioner choice	Customer can ask for a specific person. Otherwise the bot picks any available one.
Connection	A Google service account. Each practitioner shares their calendar with it once. The Calendar API is free and needs no billing.
If Google is unreachable	The bot tells the customer it will confirm the time shortly, saves a pending request, and alerts staff. No false confirmations.

5c. Feature scope: SaleSmartly-style modules plus smart extras

The system covers the same modules as SaleSmartly. Each module is delivered by Chatwoot, n8n or our admin panel, so nothing is rebuilt that already exists.

Module	What it does	Built with	When
Chat	One inbox for WhatsApp, Instagram, Messenger. Assignment, labels, private notes, canned replies. AI answers first, staff take over any time. Translation view for staff.	Chatwoot + AI service	Core
Customers	Customer list with search, filters (tag, channel, assigned staff, created date), saved views (All, My customers, Hot leads, Needs follow-up), bulk actions, CSV export. **One customer ID across channels** (matched by phone number). Contact view and customer view. Tags, tiers, notes, full chat and booking history.	Admin panel + Chatwoot contacts	Phase 6
Bookings	Calendar and list of appointments, one Google Calendar per practitioner, reschedule, cancel, attended, no-show.	Admin panel + Google Calendar	Core
Marketing	Opt-in tracking, segments by tag or intent, WhatsApp template broadcasts, reactivation of inactive customers, opt-out handling. Only to customers who agreed to receive messages.	n8n + admin panel	Phase 10
Workflow	Ready-made automations (router, reminders, follow-ups, reports) edited visually. Common ones can be switched on or off from the admin panel.	n8n	Core, extended later
Reports	Chats, leads by score, bookings, handovers, first-reply time, busiest hours, top questions, staff workload, export.	Chatwoot reports + admin panel	Phase 8
Team	Staff accounts, roles (owner, staff), teams, working hours, automatic assignment, skill-based routing (laser enquiries go to the doctor).	Chatwoot + admin panel	Core, extended later

Module	What it does	Built with	When
Settings	Connect channels, AI settings, services and prices, practitioners and calendars, FAQ, alerts.	Admin panel	Core

Extra smart and convenient features

Ideas beyond SaleSmartly's public feature list, chosen for a clinic. Wave 1 is small effort and high value; wave 2 follows once the core is live.

Extra	What it does	Wave
AI suggested replies	When staff handle a chat, the AI drafts a reply from the knowledge base. Staff review and send with one click.	1
Auto summary, tags, intent, mood	Every chat gets a short summary, tags, intent and sentiment, so staff skim instead of read.	1
Staff translation view	Staff read Chinese and English chats in their own language and reply in the customer's.	1
Click-to-chat link and QR code	A WhatsApp link and QR code for posters, receipts, website and social bios, with a tag showing where each customer came from.	1
After-hours mode	AI only at night, AI plus staff in the day. Sets customer expectations about response time.	1
Self-service changes	Customer reschedules or cancels in chat; calendar and reminders update automatically.	1
Weekly owner insights	Top questions, lost-lead reasons, busiest times, suggestions for FAQ gaps.	1
Aftercare and review requests	After a treatment: aftercare tips, a check-in a few days later, then a review request.	2
Rebooking reminders	Reminds a customer when their next session is due, based on the treatment.	2
No-show recovery	Offers a new time automatically after a missed appointment.	2
Waiting list	If a slot is full, the customer joins a list and is told if it opens up.	2
Knowledge import	Upload a price list or website page and the AI proposes FAQ entries for the owner to approve.	2
Review queue	Chats where the bot was unsure, plus suggestions to turn good staff answers into FAQ entries.	2
Voice notes and photos	Voice notes transcribed with a free open-source model. Photos attached to the handover note (the bot never judges skin conditions).	2
Deposit link	Optional payment or deposit link for high-value or no-show-prone treatments.	2
Mobile admin	The admin panel works as a phone app so the owner can manage on the go.	2

5d. Multi-business design

One installation serves many businesses. Each business is a separate tenant with its own settings, data, staff, WhatsApp number and calendars. Every table carries a business ID from the first build, because adding it later is much more expensive.

Area	How it works for many businesses
Data separation	Every record carries a business ID. Database row-level rules stop one business reading another's data, even if a query is written wrongly. Automated tests try to read across businesses.
Inbox	A Chatwoot account is created automatically for each new business (through Chatwoot's platform interface), with its own channels and staff.
Settings	AI settings, catalog, resources, FAQ, hours, alerts and lead questions are stored per business.

Area	How it works for many businesses
AI	The prompt is built per business from its own settings, languages and template. AI usage is counted per business.
n8n	One set of workflows serves everyone. The business is identified from the Chatwoot account of each message. Scheduled jobs run per business in its own time zone.
Calendars	Each business shares its Google Calendars with the platform's service account.
WhatsApp and Meta	Each business uses its own WhatsApp Business number and Meta account, so brand name and Meta fees stay with the business, and one business's mistake cannot take down the others.
Languages	Chosen per business. Chinese and English by default, others added as needed.

Roles

Role	Can do
Platform owner (you)	See and manage all businesses, plans, templates, usage, revenue and platform health. Pause or delete a business. Opens a business's data only with an audit trail.
Business owner	Everything for their own business: AI settings, catalog, staff, FAQ, bookings, reports, plan and billing.
Staff	Chats, customers and bookings of their own business only.

5e. Self-service sign-up and support for all industries

Sign-up journey

Step	What happens
1. Sign up	Public website, email and password, email verification, bot check.
2. Describe the business	Pick an industry or type a description. Optionally paste the website address or upload a price list.
3. AI drafts the setup	The AI proposes the catalog, FAQ, tone, handover rules, lead questions and hours. The owner reviews and edits. Nothing goes live without approval.
4. Calendars (if bookings)	Guided steps to share Google Calendars, with a test button.
5. Connect WhatsApp	Guided steps with screenshots and a test message. Instagram and Messenger follow.
6. Test chat	Try the assistant in each language before it talks to customers.
7. Choose a plan	Start a free trial or pick a plan.
8. Go live	Dashboard shows channel and calendar health. Help is one click away.

Aim: a working assistant within an hour once the business's Meta accounts are ready (a rough target).

How one system fits every industry

Building block	What it gives every business
Universal base	Instant replies, FAQ answers, lead capture, scoring, handover to staff. Works for any business, even with no booking.
Catalog	Services or products with name, price and description, in each language. Replaces a fixed treatment list.
Booking types	Appointment with a resource, table reservation, property viewing, class or trial, quote visit, or none. Resources can be people, rooms, agents or tables.
Custom lead questions	The owner chooses what to ask (budget, party size, location, child's age, and so on).

Building block	What it gives every business
Starter templates	A growing library. Examples below. A template pre-fills the catalog, FAQ, tone, handover rules and lead questions.
AI-assisted setup	Builds a first draft from a description, website or document, so even an industry with no template starts quickly.
Handover rules	Editable per business, with safe defaults for sensitive topics (medical, legal, payment, complaints).

Example template	What gets booked	Extra lead details	Always hand over
Aesthetic clinic	Treatment with a practitioner	Skin concern, past treatments	Medical or side-effect concerns
Dental clinic	Check-up, cleaning, treatment	Pain level, last visit, insurance	Pain, emergencies
Hair and beauty salon	Service with a stylist	Hair type, preferred stylist	Complaints
Real estate	Property viewing with an agent	Budget, area, buy or rent, timeline	Offers, legal questions
Restaurant or cafe	Table reservation (party size)	Date, party size, allergies	Large groups, events
Education or tuition	Trial class or consultation	Child's age, subject, goals	Fee disputes
Repair and home services	Quote visit or job	Location, issue, urgency	Emergencies, safety
Retail or online shop	No booking (product questions, orders)	Product, quantity, delivery area	Refunds, damaged goods

Connecting channels

Channel	At first	Later
WhatsApp	Guided manual connect: the business creates its own WhatsApp Business account and pastes its details with a test button.	One-click connect after the platform is approved by Meta as a technology provider.
Instagram and Messenger	Connect through Facebook sign-in once the platform's Meta app is approved.	Same.
Google Calendar	Share the calendar with the platform's service account.	Sign in with Google.

Meta approvals needed before public sign-up: business verification of your own company, app review for messaging permissions, public privacy policy and terms pages, and a data-deletion page. These take weeks, so they start in week 1 and run alongside the build.

5f. Plans, billing, safety and the platform-owner console

Plans and billing

Plan	Idea
Free trial	Limited days and AI replies so a business can try it. No card needed for the first trial (to decide).
Paid plans	Monthly subscription with limits on AI replies, staff seats and resources. Prices to be decided.
When a limit is reached	AI pauses and the owner is told. Messages still arrive in the inbox for staff to answer.
Meta fees	Billed by Meta directly to each business, not by the platform.
Payment	Stripe or a local payment provider (they charge a fee per payment). Usage and invoices are shown to the owner.

Safety and compliance

Area	What we do
Abuse prevention	Email verification, bot check, rate limits, tight limits on new accounts, block list, AI cost caps per business.
Acceptable use	Rules against spam and Meta-prohibited industries. Every business agrees. A kill switch pauses one business instantly.
Data protection	Encrypted tokens and credentials, two-step login for owners, audit logs, backups, and export and deletion of a business's data on request.
Legal pages	Terms of service, privacy policy, data processing terms, and an AI disclosure and consent line in chats. Have a lawyer review them for your country.
Support access	The platform owner views a business's data only with a recorded reason.

Platform-owner console

A list of all businesses with status and plan, usage and revenue, channel and calendar health, quick pause and delete, template management, plan changes, announcements, and alerts when a business has problems.

Honest hosting note: an invite-only beta with a handful of businesses can run on a free server. Open sign-up to the public needs a stable paid server with backups and monitoring, a real domain name, and an email-sending service. Expect roughly $30 to $100 a month at the start, funded by plan revenue. Moving is straightforward because everything is plain Docker.

6. How the AI assistant behaves

Language

- Replies in the language of the customer's latest message, including mixed Chinese and English.
- Matches Simplified or Traditional Chinese to what the customer writes.
- Uses fixed treatment names from the business file, so names and prices never drift.

What it can do (tools)

Tool	Purpose
check_availability	Real open time slots for a service on a date, from Google Calendar free time of the practitioners who can do it, clinic hours and treatment length.
create_booking	Re-checks the slot, saves the appointment and creates the calendar event. Asks for name and phone first.
list_my_bookings / cancel_booking	Lets the customer see or cancel their own booking.
save_lead	Stores name, phone, interest, preferred time, language and a hot / warm / cold score.
handover_to_human	Marks the chat as staff-owned and passes a short summary.

Guardrails

- Answers only from the clinic knowledge. Never invents prices, promotions, or availability.
- No medical diagnosis, no guaranteed results. Medical questions go to staff.
- Tells the customer on the first message that it is a virtual assistant.
- Treats customer text as untrusted. Cannot be talked into changing prices or revealing its rules.
- If the AI service fails, the customer gets a polite message and staff are alerted, never silence.

Handover to a person happens when

- The customer asks for a human.
- Complaint, anger, refund or payment problem.
- Medical concern, allergy, side effect, pregnancy or similar.

- The question is outside the knowledge base, or the AI is unsure.
- The customer sends an image or voice note the assistant cannot read and it matters.

On handover n8n adds a private note with the AI summary, sets the conversation to Open, adds the label needs-human, and sends a Telegram alert. The bot stays silent until staff type /bot.

7. First template: aesthetic clinic

Typical customer questions the assistant handles: treatment list and price ranges, how long a treatment takes, downtime and aftercare, who it is suitable for, location and parking, opening hours, current promotions, first-visit steps, and booking, rescheduling or cancelling.

Business content (services, prices, practitioners, hours, FAQ) is entered in the admin panel and stored in the database. The kit ships with bilingual sample content, loaded on first start.

All sample content will be clearly marked as sample and must be replaced with the real clinic's information and reviewed by the clinic before going live.

8. Meta platform rules to plan for

- **WhatsApp:** needs a Meta Business account and a phone number that is not already used on the normal WhatsApp app. Free-form replies are allowed only within 24 hours of the customer's last message. Outside that window only pre-approved templates can be sent.
- **Templates:** create English and Chinese versions of the appointment reminder and get them approved.
- **Instagram and Messenger:** the Meta app needs permissions approved through app review. This can take days to weeks, so start early.
- **Public HTTPS address** for the Chatwoot webhook (section 4).
- **Privacy:** short consent line when collecting name and phone, and an easy way to opt out.
- Check Meta's current pricing and rules before launch, because they change.

9. Data stored

Table	Holds
settings / services / practitioners	Everything edited in the admin panel: AI settings, FAQ text, treatments, practitioners and their calendar IDs, hours.
tenants	One row per business: name, template, time zone, languages, Chatwoot account, Meta and calendar connections, plan and limits. Every other table carries the business ID.
usage	Messages, AI usage and bookings per business per day, for reports and limits.
plans / subscriptions	Plan, limits, trial dates, payment status and invoices per business.
catalog / resources / lead questions	Per-business products or services, bookable resources, and custom lead questions.
audit_log	Who did what: sign-ins, support access, pauses, deletions.
admin_users	Panel logins (hashed passwords) and roles: platform owner, business owner, staff.
conversations	Contact name, phone, channel, language, bot or human mode, handover reason and summary, time of last customer message.
messages	Recent chat history used as AI memory. Duplicate webhook deliveries are ignored.
leads	Name, phone, interest, preferred time, score, status, notes.
bookings	Customer, service, practitioner, Google event ID, start and end time, status, language, reminder flags.

Backups: a scheduled database dump kept off the machine (for example a free cloud drive). Secrets stay in a .env file that is never committed.

10. Build phases

Phase	Deliverable	Rough time
0. Accounts and Meta approvals	Meta Business verification for your company, WhatsApp number, Meta app review request, domain, server, Anthropic key, Telegram bot. Approvals run alongside the build.	set-up 1 week, approvals take weeks
1. Stack and multi-business base	Docker Compose: Chatwoot, n8n, Postgres, AI service, tunnel. Business ID and row-level rules on every table.	4-5 days
2. Core bot	Bilingual answers from the knowledge base, WhatsApp reply loop	3-4 days
3. Leads and handover	Lead scoring, private note, staff alert, give-back command	2-3 days
4. Booking and Google Calendar	Generic booking types and resources, availability, cancel, reminders	6-8 days
5. Business admin panel	Dashboard, AI settings, catalog, resources, FAQ, bookings, test chat, users	6-8 days
6. Customers module	One customer ID across channels, tags and tiers, saved views, bulk actions, export	4-5 days
7. Self-service sign-up	Public site, sign-up and verification, AI-assisted setup, guided channel connect, trial	8-10 days
8. Plans, limits and billing	Plans, usage limits, payment provider, invoices	5-7 days
9. Platform-owner console and safety	Business list, pause and delete, templates, abuse protection, legal pages, audit log	5-7 days
10. Channels	Instagram and Messenger connect	1-2 days plus Meta review
11. Reports, hardening, polish	Reports, backups, health alerts, security and isolation tests, staff help guides	6-8 days
12. Smart extras, wave 1	Suggested replies, auto summary and tags, translation view, click-to-chat link, after-hours mode, self-service changes, weekly insights	5-7 days
13. Marketing and workflows	Opt-in, segments, template broadcasts, reactivation, aftercare and rebooking messages	5-7 days
14. Wave 2 and one-click WhatsApp	Remaining extras and one-click WhatsApp sign-up after Meta approval	as needed

Milestones (rough estimates): an invite-only beta on free hosting (phases 0 to 5, 7, 10 and the core of 11) in about 10 to 12 weeks. Open public self-service (adding phases 8 and 9, Meta approvals and a paid server) in about 16 to 20 weeks. Everything including smart extras and marketing in about 20 to 24 weeks. Meta approvals are the main schedule risk.

11. Running costs

Item	Cost	Note
Server / VM	$0	Always-free VM or an existing machine.
Google Calendar API	$0	Free for this volume. No billing account needed.
Server for public sign-up	About $20 to $80 a month	Free hosting is fine for development and a small beta. Open sign-up needs a stable paid server with backups.
Domain and email sending	About $10 a year, plus a free or low email tier	Needed for the sign-up site, Meta review pages and verification emails.
Payment fees	Per payment	Charged by the payment provider on each subscription.

Item	Cost	Note
Chatwoot, n8n, Postgres, tunnel	$0	Open source or free tiers.
WhatsApp Cloud API	Low	Customer-started chats are inside the free service window; template messages (for example reminders outside the window) are charged by Meta. Confirm current rates.
Claude API (Haiku 5.5)	Pay per use, small	List price $0.10 per million input tokens and $0.50 per million output tokens. A typical chat is estimated at well under one US cent, so roughly a few dollars per 1,000 chats. Prompt caching lowers it further. This is the one unavoidable cost, because there is no free hosted model of this quality.

The model is set by one setting, so it can be raised to a stronger model later if answer quality needs it. Smart extras (summaries, suggested replies) add more AI calls per chat, so usage grows a little.

12. Risks and how we reduce them

Risk	Mitigation
Machine goes offline	Use an always-on VM. Health check alerts to Telegram.
Tunnel URL changes	Use ngrok static domain, not the quick tunnel.
AI gives a wrong price or claim	Answer only from the knowledge base, strict rules, staff review the sample content, review early chats.
Customer stuck with bot	Always-available human option, automatic handover rules.
Bot and staff both reply	Per-conversation mode: bot is silent whenever staff own the chat.
Meta approval delays	Start verification and app review in week 1. Launch WhatsApp first.
Google calendar not shared or token revoked	Admin dashboard shows calendar connection status per practitioner. Alert to staff if a check fails.
Admin panel exposed to the internet	Not published through the public tunnel. Strong passwords, roles, access over a private secure link.
One business affecting another	Business ID on every record, database row-level rules, per-business tokens and rate limits, and tests that try to read across businesses.
Each new business needs Meta set-up	Start Meta verification early for each client. Use the onboarding wizard checklist so nothing is missed.
AI cost differs by business	Usage counted per business, optional limits, and a monthly report.
Meta approvals delay public launch	Start verification and app review in week 1. Run an invite-only beta meanwhile.
Abuse or spam sign-ups	Email verification, bot check, rate limits, new-account caps, kill switch.
A business breaks WhatsApp or Instagram rules	Acceptable-use terms, per-business Meta accounts so others are unaffected, ability to pause a business.
Legal and data-protection duties	Terms, privacy policy, deletion and export, and a lawyer review for your country.
Support workload	Guided onboarding, built-in help, and the platform's own assistant answering common questions.
Free tier limits change	Everything is plain Docker, so it can move to any other machine.

13. Reference: SaleSmartly

SaleSmartly (salesmartly.com) is a ready-made cloud platform for a very similar job: one inbox for many chat channels, AI agents with human takeover, a customer database, and broadcasts. Its app is behind a login, so I could not view its screens. This comparison uses its public website and one third-party review. Vendor claims are unverified, and no screen layouts were copied.

SaleSmartly module	What it offers (public information)	In our plan
Multi-channel inbox	Lists 14 channels, including WhatsApp, Messenger, Instagram, Telegram, LINE, WeChat, email and website chat. Keeps conversation history. Real-time translation.	Chatwoot inbox. WhatsApp, Instagram and Messenger first. Chatwoot also has website chat, Telegram, LINE and email. WeChat is not included.
AI agents + human	AI first reply, remembers customer needs, judges intent, follows up proactively, hands key chats to a person.	Same idea: AI with per-customer memory, lead scoring, handover rules. Proactive follow-up is workflow 4.
SCRM (customer records)	Company-owned customer data, tags and tiers, automatic assignment, routing links and QR codes, analytics.	Leads and bookings in the admin panel. Add tags and tiers, auto-assign handovers to staff, and a click-to-chat WhatsApp link and QR code.
Re-marketing	Targeted WhatsApp bulk messages by tag or intent, reactivation, automation.	Optional later: n8n campaign using approved templates, only to customers who opted in.
Attribution	Links chats to ad campaigns, Meta Conversion API.	Optional later.

Build with our plan or use SaleSmartly?

	Our plan (self-hosted)	SaleSmartly (cloud service)
Running cost	$0 hosting, plus small AI usage cost.	Monthly subscription. One third-party review quotes about $15.90 per month for a basic plan (2 agents, 10 accounts) and mentions a free plan. Check current prices.
Time to launch	Beta in about 10 to 12 weeks, public sign-up in 16 to 20 (section 10).	Days, because it is already built.
Data and control	Everything on your own server. Rules written exactly for your clinic.	Data held by the vendor. Customising limited to what the plan offers.
Booking with one Google Calendar per practitioner	Built to your rules, with calendar check before every booking.	Not confirmed in what I could read. Ask the vendor before relying on it.
Upkeep	You (or I) maintain it.	Vendor maintains it.
Both need	A verified Meta Business account and a WhatsApp number. Meta fees apply either way.	

Ideas we adopt: staff translation view (read Chinese chats in English and the reverse), customer tags and tiers, automatic assignment of handovers, a click-to-chat link and QR code for posters and the website, and customer memory. If speed matters more than control, trying SaleSmartly's free plan in parallel is a reasonable way to compare. Section 5c turns these ideas into our feature list.

Sources: https://www.salesmartly.com/ and https://gptbots.ai/blog/salemartly-review

14. What I need from you to start building

- Approval of this plan, or changes you want.
- Platform name, domain name and the country that billing starts in.
- Plan structure: free trial length, paid plan limits and prices, and which payment provider to use.
- The legal entity that will own the Meta business verification and sign the terms.
- Agreement to start with an invite-only beta before opening sign-up to the public.
- Where it will run: always-free cloud VM (recommended) or an office machine.
- Time zone and currency for the business (the sample assumes Malaysia, ringgit).
- List of practitioners (names, what each can do, working hours) and a Google account that owns their calendars.

- When ready: Meta Business account, a WhatsApp number, an Anthropic API key, and a Telegram bot for alerts.

On approval, the next deliverable is the full starter kit in the aichat repository: Docker Compose stack, database schema, AI service with admin panel and Google Calendar sync, importable n8n workflows, bilingual sample content, and a step-by-step setup guide.